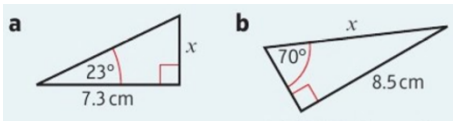
Chapter 9 · Trigonometric ratios

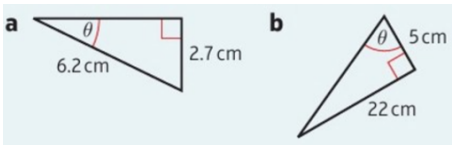
Starters 2

1. Find the marked lengths.



1(a,b)

2. Find the marked angles.



2(a,b)

3. Given $f(x) = x^2 + 3x$, sketch each graph. All four are upward-opening parabolas.

a.

$y = f(x)$

a

b.

$y = f(x + 2)$

b

c.

$y = f(x) - 3$

c

d.

$y = f(\frac{1}{2}x)$

d

Chapter Overview 3

1. Sine and cosine rules

Find missing sides and angles in non-right-angled triangles.

2. Areas of triangles

Use two sides and the included angle.

3. Trigonometric graphs

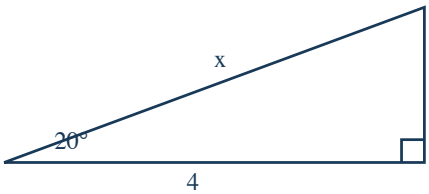
Sketch sine, cosine and tangent, then apply transformations.

RECAP: Right-Angled Trigonometry 4

Meaning. What does $\sin\theta = \text{opposite}/\text{hypotenuse}$ mean?

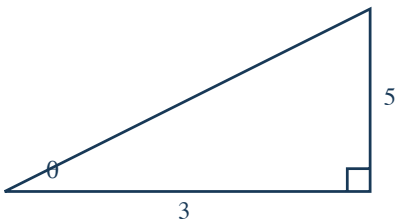
Meaning

Find x.



x

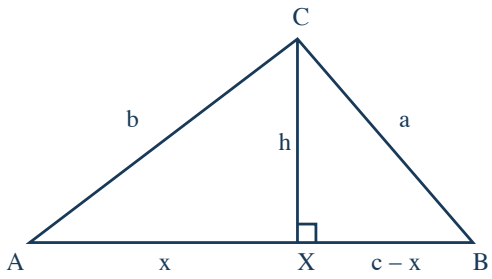
Find θ.



θ

Write the ratio equation first. Then rearrange for a length, or apply the inverse trigonometric function for an angle. Use degree mode.

Proof of Cosine Rule 5



Drop a perpendicular CX to AB. Let AX=x and AB=c.

Use Pythagoras in the two right-angled triangles.

1

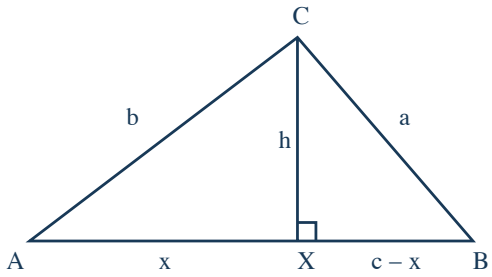
2. Eliminate h^2 and expand.

2

3. Use the angle at A.

3

Proof of Sine Rule 6



The same altitude h belongs to both right-angled triangles.

1. Express h in two ways.

1

2. Rearrange.

2

3. Repeat with another altitude.

3

Just for your interest. . . 7

Why does “cosine” contain the word “sine”?

Complementary angles add to 90°.

Supplementary angles add to 180°.

1. The acute angles of a right-angled triangle are 50° and 40°. Let x be the side opposite 40° and z the hypotenuse. Compare cosine of 50° with sine of 40°.

1

2. Generalise this relationship.

2

OVERVIEW: Finding missing sides and angles 8

Choose the rule from the information available. Lower-case side labels are opposite the matching capital-letter angles.

1. A known opposite side–angle pair and another side or angle.

1

2. Two sides and the included angle; find the third side.

2

3. All three sides; find an angle.

3

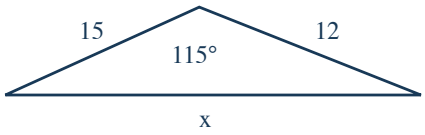
4. Two sides and a non-included angle; find another side.

4

Cosine Rule 9

$a^2 = b^2 + c^2 - 2bc\cos A$

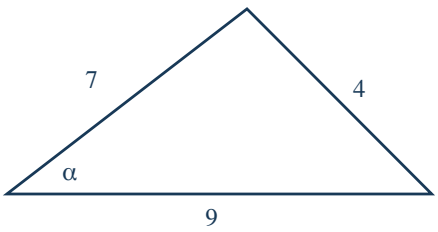
A is the angle between b and c, opposite a.



Find x.

x

Dealing with Missing Angles 10



Find α, opposite the side of length 4.

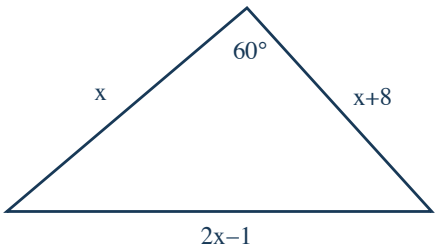
1

2

Alternatively use $\cos A = (b^2 + c^2 - a^2)/(2bc)$. Only the product $2bc$ is multiplied by $\cos A$, not the entire right-hand side.

Harder Ones

1. Determine x.



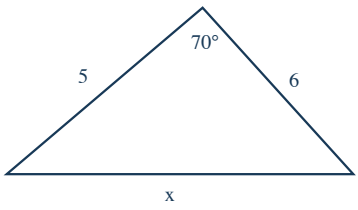
1

2. Coastguard station B is 8 km on a bearing of 060˚ from A. Ship C is 4.8 km on a bearing of 018˚ from A. Find BC.

2

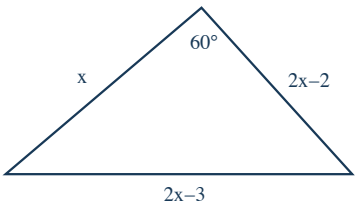
Test Your Understanding

1. Find x.



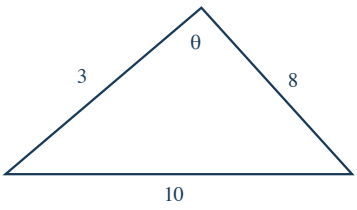
1

2. Find x.



2

3. Find θ.



3

An obtuse angle has a negative cosine.

8. Triangle PQR has PQ=2 cm, PR=3 cm and QR=4 cm. Show that $\cos P = -\frac{1}{4}$.

8

11. The side lengths of a triangle are in the ratio 2:3:4. Calculate the largest angle.

11

17. Ships A, B and C lie in the same horizontal plane. B is 50 km due north of A. C is 70 km from A on a bearing of 020°.

a. Find BC to 3 significant figures.

a

b. Find the bearing of C from B.

b

Extension 15

[STEP I 2009 Q4(i)] A triangle has sides p−q, p and p+q, where p>q>0. Its largest and smallest angles are α and β. Show that

$4(1-\cos\alpha)(1-\cos\beta)=\cos\alpha+\cos\beta$

1. Apply the cosine rule opposite the longest and shortest sides.

1

2. Compare both sides of the identity.

2

The Sine Rule 16

Compare side length divided by the sine of its opposite angle.

1. Opposite 85°: side 10; opposite 30°: side 5.02; opposite 65°: side 9.10.

1

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$

Use the sine rule when you know a complete opposite side–angle pair and one part of another pair. Small discrepancies above come from rounded side lengths.

Examples 17

1. In a triangle, side 8 is opposite 45° and side x is opposite 85°. Find x.

1

2. In a triangle, side 8 is opposite 30° and side x is opposite 100°. Find x.

2

Match each side to its opposite angle. The remaining angles are 50° in both examples.

Examples 18

For a missing angle it is convenient to use $\sin A/a = \sin B/b$.

3. Side 6 is opposite 85°, and side 5 is opposite θ. Find θ.

3

4. Side 10 is opposite 126°, and side 8 is opposite θ. Find θ.

4

8. B is 6 km on a bearing of 020° from A. C is on a bearing of 055° from A and 120° from B. Find AC and BC.

Angles. Find the interior angles.

Angles

a. Find AC.

a

b. Find BC.

b

11. In triangle ABC, AB=x cm, BC=(4−x) cm, angle BAC=y and angle BCA=30°. Given $\sin y = 1/\sqrt{2}$, show that $x = 4(\sqrt{2} - 1)$.

1. Use the sine rule.

1

2. Rationalise the denominator.

2

12. A surveyor measures the angles of elevation of the top of a building from two points 15 m apart, on level ground in a straight line towards the building. The angles are 40° at the farther point and 52° at the nearer point.

a. Find the height of the building.

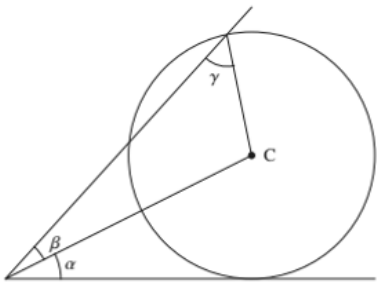
a

b. State an assumption.

b

Extension 22

[MAT 2011 1E] The circle has centre C. Which relationship is correct?



A: $\cos\alpha = \sin(\beta + \gamma)$

B: $\sin\beta = \sin\alpha\sin\gamma$

C: $\sin\beta(1 - \cos\alpha) = \sin\gamma$

D: $\sin(\alpha + \beta) = \cos\gamma\sin\alpha$

Answer

The ‘Ambiguous Case’ 23

AB=4, AC=3 and angle ABC=44°. Find the possible values of angle ACB.

1. Apply the sine rule.

1

2. Find both possible angles.

2

The ambiguous case: $\sin\theta = \sin(180° - \theta)$. The calculator returns the principal inverse-sine value; the sine equation may also have a supplementary solution. Check the triangle and any stated conditions.

Test Your Understanding 24

A triangle has side 10 opposite an obtuse angle θ, and side 5 opposite 20°. The remaining side is x.

1. Find θ.

1

2. Find x.

2

4. Triangle ABC has AB=4 cm, BC=6 cm and angle ACB=36°. Show that one possible value of angle ABC is 25.8° (3 s.f.), and use this case to find AC.

a. Find the relevant angle.

a

b. Find AC using unrounded angles.

b

5. Two triangles ABC have AB=4.5 cm, BC=6.8 cm and angle ACB=30°. Find the largest angle in each triangle.

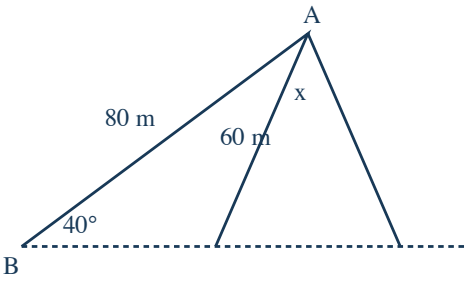
1. Find both possible values of A.

1

2. Identify the largest angles.

2

6. Crane arm AB is 80 m long and makes an angle of 40° with the horizontal at B. A wrecking ball hangs on a 60 m cable from A.



Find the angle x through which the cable rotates between the two positions where the ball is level with B.

a

State a modelling assumption.

b

Area of Non Right-Angled Triangles 28

$$\text{Area} = \frac{1}{2}ab\sin C$$

C is the included angle between sides a and b.

Example. Find the area of a triangle with sides 3 cm and 7 cm and included angle 59°.

Example

Test Your Understanding 29

1. A triangle has sides x and x+3, included angle 30°, and area 10. Find x.

1

2. A triangle has sides 5 and 6, included obtuse angle θ, and area 10. Find θ.

2

Exercise 9D – Y1 pages 186-187 30

4. Triangle ABC has BC=(x+2) cm, AC=x cm, angle BCA=150° and area 5 cm^2. Find x to 3 significant figures.

4

5. Triangle PQR has PQ=(x+2) cm, PR=(5−x) cm and angle QPR=30°. Its area is A cm².

a. Show that $A = \frac{1}{4}(10 + 3x - x^2)$.

a

b. Find the maximum area and the corresponding x.

b

6. Triangle ABC has AB=x cm, AC=(5+x) cm, angle BAC=150° and area 3¾ cm².

a. Show that $x^2 + 5x - 15 = 0$.

a

b. Calculate x to 3 significant figures.

b

Sin or cosine rule? 33

Choose sine or cosine rule, and explain why. Check that the measurements can form a triangle.

1. Two sides are 3 and 5; their included angle is 100°. Find the opposite side.

1

2. Two angles are 40° and 45°, and one side is 3. Find another side.

2

3. The three proposed side lengths are 2, 3 and 5. Find an angle.

3

4. Two sides are 3 and 5, with 40° opposite side 3. Find the remaining side.

4

Using sine rule twice 34

Two sides of a triangle are 4 and 3. The angle opposite side 3 is 32°. Let the remaining side be x.

Cosine route. What equation would the cosine rule give?

Cosine route

For side–side–angle information, there may be two triangles. A diagram alone should not be used to discard the other solution unless an angle condition is stated.

Using sine rule twice 35

Continue: side 3 is opposite 32˚, side 4 is opposite A, and the remaining side is x.

1. Use the sine rule for A.

1

2. Find the remaining angle B.

2

3. Use the sine rule again.

3

Test Your Understanding 36

1. A triangle has side 9 opposite 61˚ and another side 10. Find the possible lengths of the remaining side y.

1

2. A triangle has side 4 opposite 53˚ and another side 3. Find its area.

2

Problem Solving With Sine/Cosine Rule 37

Four mobile phone masts form convex quadrilateral ABCD. BC=75 m, CD=80 m, angle BCD=55° and angle ADC=140°. Masts must be at least 70 m apart, and AD is the minimum permitted distance.

a. Find AB.

a

b. Find angle BAD.

b

c. Find the enclosed area.

c

Exercise 9E – Y1 page 189-191 38

12. Triangle ABC has perimeter 15 cm, AB=7 cm and angle BAC=60°. Find AC, BC and the area.

Sides. Let AC=x, so BC=8−x.

Sides

Area. Use the included angle.

Area

13. Triangle ABC has AB=14 cm, BC=12 cm and CA=15 cm.

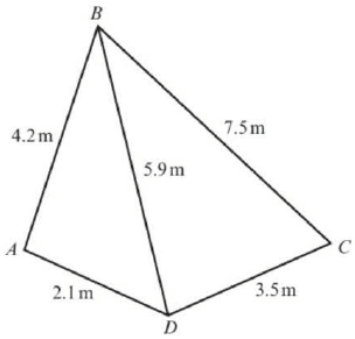
a. Find angle C to 3 significant figures.

a

b. Find the area to 3 significant figures.

b

14. A flower bed is the quadrilateral shown. All lengths are in metres.



a. Find angles DAB and BCD.

a

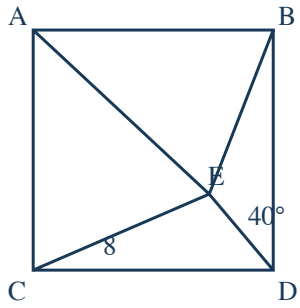
b. Find the total area.

b

c. Find diagonal AC.

c

15. ABCD is a square of side 10 cm, with C bottom-left and D bottom-right. E lies inside the square, CE=8 cm and angle BDE=40°. Angle CED is obtuse. Find the area of triangle ABE.



1. Find the height of E above CD.

1

2. Find the area of triangle ABE.

2

Extension Problem 1 42

[AEA 2009 Q5(a)] Triangle ABC has sides BC=a, AC=b, AB=c, with a<b<c. Angles A, B, C form an arithmetic sequence. Given $a = 2$ and $\sin A = \sqrt{15}/5$:

i. Show that the area is $ac\sqrt{3}/4$.

i

ii. Find b.

ii

iii. Find c.

iii

[STEP I 2006 Q8] O=(0,0,0), A=(a,0,0), B=(0,b,0), C=(0,0,c), where a,b,c>0. Let angle ACB=θ and let d be the perpendicular distance from O to plane ABC.

i. Find the volume of tetrahedron OABC.

i

ii. Find cos θ and the area Δ of ABC.

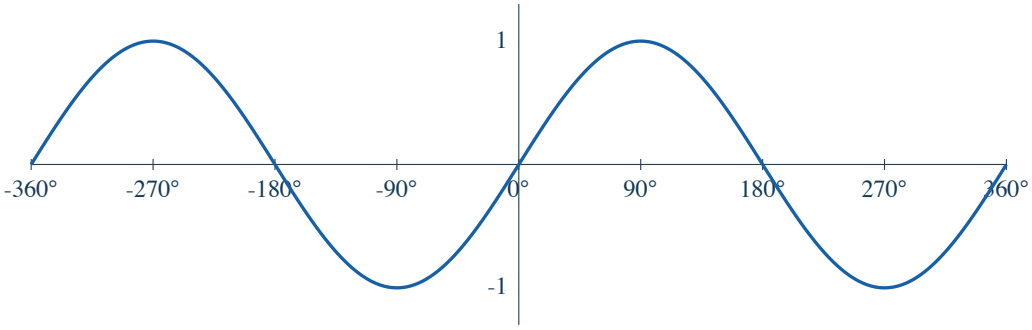
ii

iii. Show that $d^{-2} = a^{-2} + b^{-2} + c^{-2}$.

iii

Sin Graph 44

$y = \sin x$

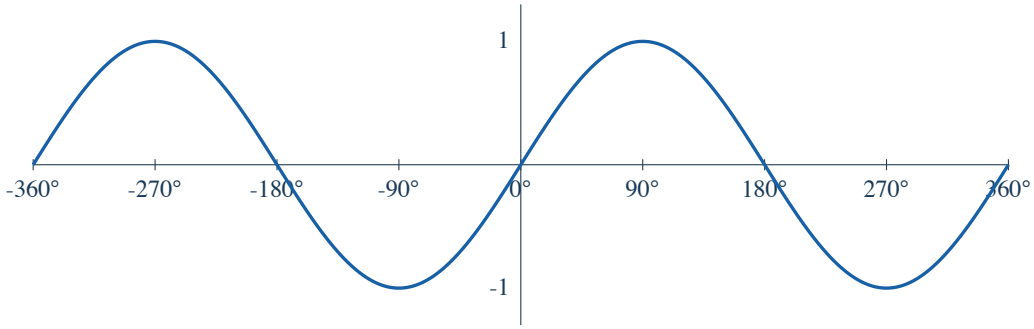


Features. State the period, range and zeros.

Features

Sin Graph 45

Given $\sin 30^\circ = \frac{1}{2}$, use symmetry of the sine graph.



a.
$\sin 150^\circ$

a

b.
$\sin(-30^\circ)$

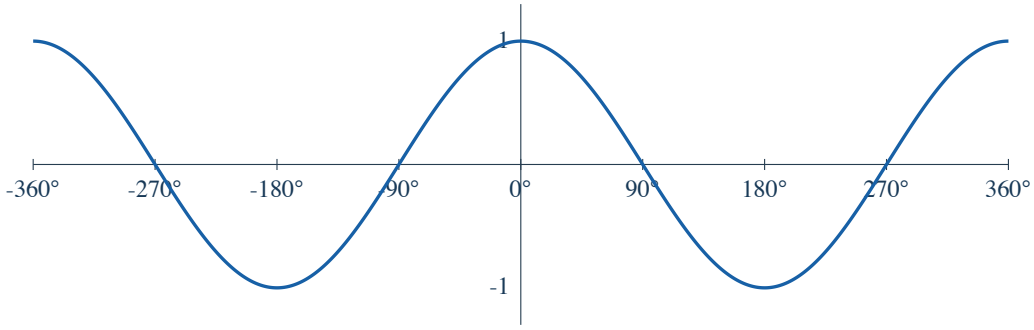
b

c.
$\sin 210^\circ$

c

Cos Graph 46

$y = \cos x$

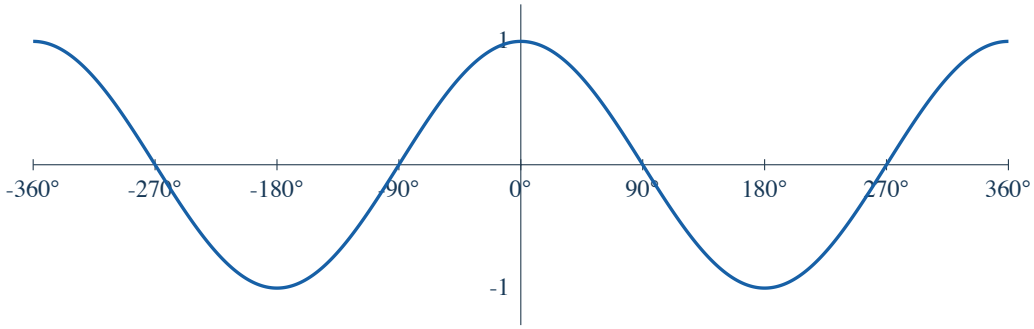


Features. State the period, range and zeros.

Features

Cos Graph 47

Given $\cos 60^\circ = \frac{1}{2}$, use symmetry of the cosine graph.



a.
$\cos 120^\circ$

b.
$\cos(-60^\circ)$

c.
$\cos 240^\circ$

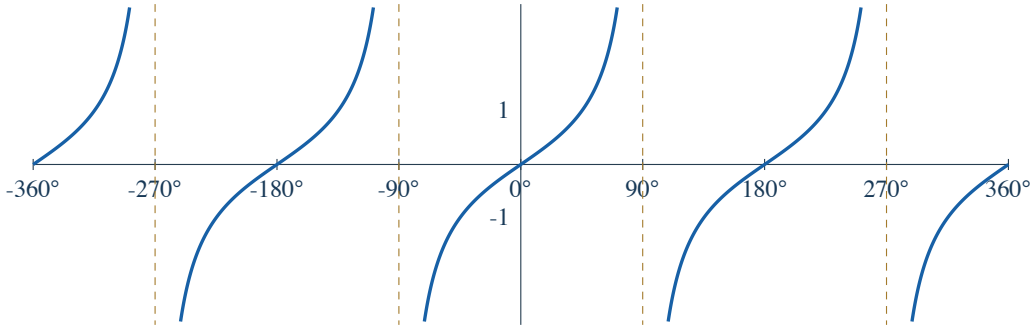
a

b

c

Tan Graph 48

$y = \tan x$

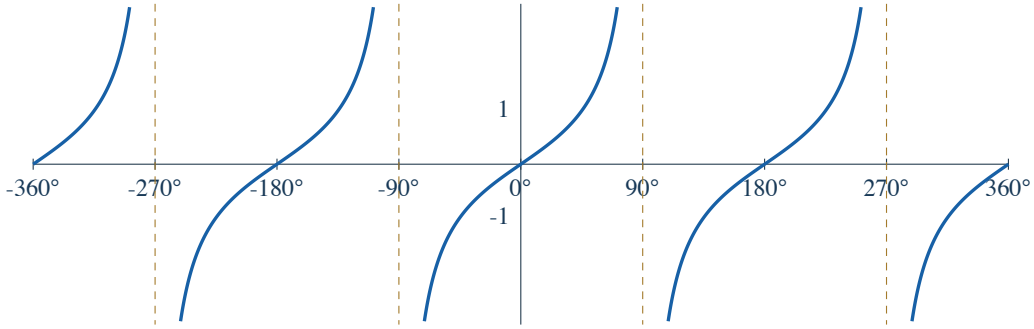


Features. State the period, range, zeros and asymptotes.

Features

Tan Graph

Given $\tan 30^\circ = 1/\sqrt{3}$, use symmetry of the tangent graph.



a.

$\tan(-30^\circ)$

a

b.

$\tan 150^\circ$

b

Transforming Trigonometric Graphs

Use the same transformation rules as for other functions: changes outside affect outputs; changes inside affect inputs.

1. $y = 4\sin x, \quad 0^\circ \le x \le 360^\circ$

1

1 features

2. $y = \cos(x + 45^\circ), \quad 0^\circ \le x \le 360^\circ$

2

2 features

1. $y = -\tan x, \quad 0^\circ \le x \le 360^\circ$





2. $y = \sin(x/2), \quad 0^\circ \le x \le 360^\circ$





Sketch each graph on the stated interval.

1. $y = \cos\theta,\ -180^\circ \le \theta \le 180^\circ$



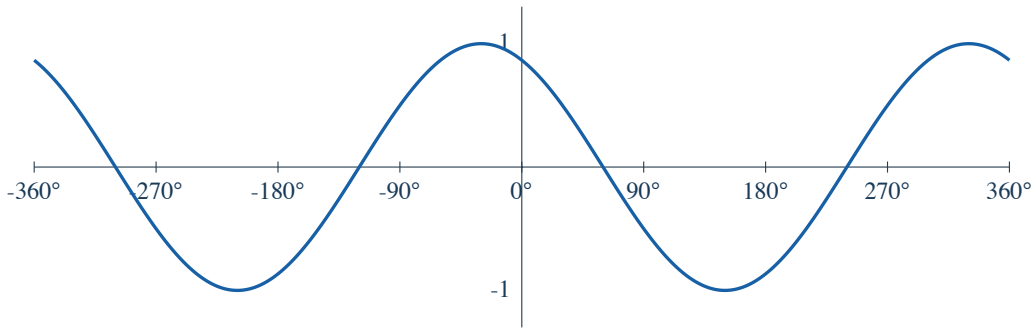
2. $y = \tan\theta,\ -180^\circ \le \theta \le 180^\circ$



3. $y = \sin\theta,\ -90^\circ \le \theta \le 270^\circ$



7. $y = \cos(x + 30^\circ), \quad -360^\circ \leq x \leq 360^\circ$



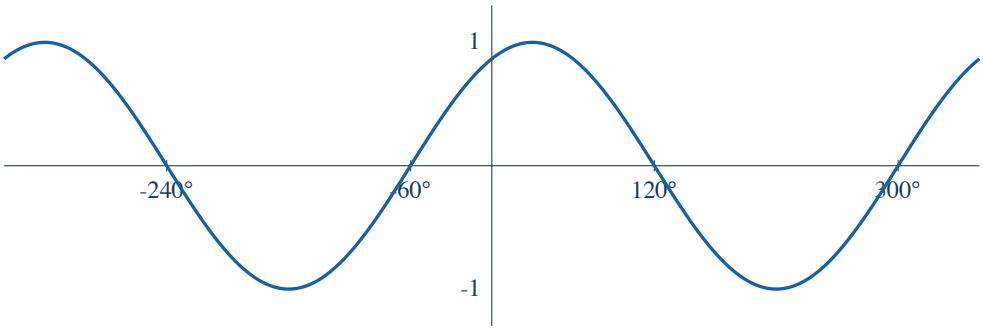
a. Find the x-axis intersections.

a

b. Find the y-axis intersection.

b

8. The graph has equation $y = \sin(x + k)$, where k is a constant (angles in degrees).



a. Give one possible value of k.

a

b. Is there more than one possible value?

b

9. The relative water depth of a rock pool is modelled by $y = \sin(30t)^\circ$, where t is time in hours and $0 \leq t \leq 6$. t=0 represents midday.

a. Sketch the function.

a

b. During what times is the pool at least half full?

b

Extension 56

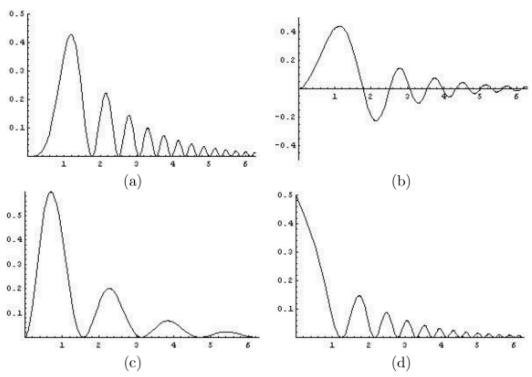
1. [MAT 2013 1B] Reflect $y = \sin x$ first in x=π, then in y=2. Find the resulting equation.

1

2. [MAT 2011 1D] What fraction of $0^\circ \leq x \leq 360^\circ$ satisfies at least one of $\sin x \geq \frac{1}{2}$ or $\sin 2x \geq \frac{1}{2}$?

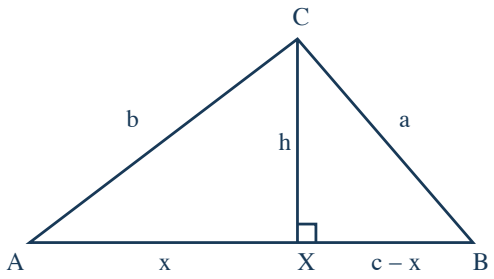
2

3. [MAT 2007 1G] Choose the sketch of $y = 2^{-x}\sin^2(x^2)$.



3

APPENDIX :: Proof of Cosine Rule



Drop a perpendicular CX to AB. Let AX=x and AB=c.

Use Pythagoras in the two right-angled triangles.

1

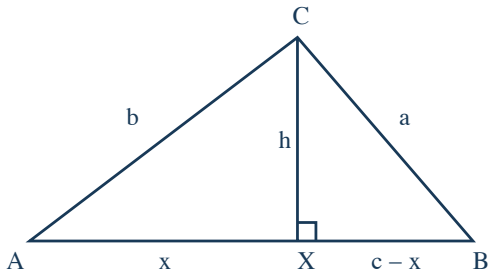
2. Eliminate h^2 and expand.

2

3. Use the angle at A.

3

APPENDIX :: Proof of Sine Rule



The same altitude h belongs to both right-angled triangles.

1. Express h in two ways.

1

2. Rearrange.

2

3. Repeat with another altitude.

3